

Ma, Pingchuan

PhD Student
CompVis Group
LMU, Munich

Email: [pingchuan_m\[at\]hotmail.com](mailto:pingchuan_m[at]hotmail.com)
[Github](#); [Google Scholar](#)

ABOUT ME

I am a final year [MCML](#) PhD student advised by [Björn Ommer](#) in the Computer Vision & Learning Group at [LMU](#) Munich.

My current research spans three main directions: 1) *multimodal representation learning*, where I study alignment and distributional relationships across modalities, with a particular focus on vision-language models; 2) *generative modeling*, where I work on improving the training and inference of diffusion and flow-based models; and 3) *model repurposing*, in which I explore using generative models for tasks beyond image synthesis, such as analyzing the evolution of art over time or learning robust, disentangled representations. These directions bridge my main research interests in representation learning and generative AI.

Earlier in my research, I focused on deep metric learning, developing robust and generalizable representations, as well as style transfer, studying how to define style from a computer vision perspective and applying it for image editing.

SELECTED PUBLICATIONS

- [Denoising, Fast and Slow: Difficulty-Aware Adaptive Sampling for Image Generation](#), in CVPR, 2026. Schusterbauer, Johannes, Ming Gui, Yusong Li, **Pingchuan Ma**, Felix Krause, and Björn Ommer.
- [Diffusion Models and Representation Learning: A Survey](#), in TPAMI, 2026. Michael Fuest, **Pingchuan Ma**, Ming Gui, Johannes Schusterbauer, Vincent Tao Hu, Björn Ommer.
- [Stochastic Interpolants for Revealing Stylistic Flows across the History of Art](#), in ICCV, 2025. **Pingchuan Ma***, Ming Gui*, Johannes Schusterbauer, Xiaopei Yang, Olga Grebenkova, Vincent Tao Hu, Björn Ommer.
- [SCFlow: Implicitly Learning Style and Content Disentanglement with Flow Models](#), in ICCV, 2025. **Pingchuan Ma***, Xiaopei Yang*, Ming Gui, Yusong Li, Felix Krause, Johannes Schusterbauer, Björn Ommer.
- [Does VLM Classification Benefit from LLM Description Semantics?](#) in AAAI, 2025. **Pingchuan Ma***, Lennart Rietdorf*, Dmytro Kotovenko, Vincent Tao Hu, and Björn Ommer. We got an invited talk at [Imageomics-AAAI-25](#)
- [DepthFM: Fast Monocular Depth Estimation with Flow Matching](#), in AAAI (Oral), 2025. Ming Gui, Johannes Schusterbauer, Ulrich Prestel, **Pingchuan Ma**, Dmytro Kotovenko, Olga Grebenkova, Stefan A. Baumann, Vincent Tao Hu, Björn Ommer.
- [Boosting Latent Diffusion with Flow Matching](#), in ECCV (Oral), 2024. Johannes Schusterbauer*, Ming Gui*, **Pingchuan Ma***, Nick Stracke, Stefan A. Baumann, and Björn Ommer.
- [Cross-Image-Attention for Conditional Embeddings in Deep Metric Learning](#), in CVPR 2023. Dmytro Kotovenko, **Pingchuan Ma**, Timo Milbich and Björn Ommer.
- [Improving Deep Metric Learning by Divide and Conquer](#), in TPAMI, 2021. Artsiom Sanakoyeu, **Pingchuan Ma**, Vadim Tschernezki, and Björn Ommer.
- [A Content Transformation Block For Image Style Transfer](#), in CVPR, 2019. Dmytro Kotovenko, Artsiom Sanakoyeu, **Pingchuan Ma**, Sabine Lang and Björn Ommer

EDUCATION

PhD Student: Computer Vision & Learning Group at the University of Munich, Germany	Nov. 2021 – present
---	----------------------------

Main focus: Representation Learning, Diffusion Models, Flow Matching Models

Supervisor: Prof. Dr. Björn Ommer

PhD Student: Computer Vision Group at Heidelberg University, Germany	Jul. 2021 – Oct. 2021
---	------------------------------

Supervisor: Prof. Dr. Björn Ommer

Master of Science: Heidelberg University, Germany	Sep. 2017 – Jan. 2021
--	------------------------------

Major: Applied Computer Science

Grade (German system): 1.5

Bachelor of Engineering: Guangzhou University of Chinese Medicine, China	Sep. 2011 – Jun. 2015
---	------------------------------

Major: Computer Science

GPA (US 4.0 scale): 3.35 Ranking: top 15%

EXPERIENCE

Research Assistant in Computer Vision & Learning Group, University of Munich

Oct. 2021 – present

Focus: Multimodal Representation Learning & Generative Models

Advised by Prof. Björn Ommer

In addition to the applied projects listed below, I focus on generalizable, academic-oriented research on multimodal representation learning and generative modeling. For more details, see my About Me section and selected publications.

Industrial Collaboration with Bayer AG

Jul. 2023 – Jun. 2026

Project 1: This project focuses on adapting SOTA generative models to synthesize realistic plant images, both in controlled laboratory environments and in the wild. The goal is to conditionally generate images of plants under specific treatments, thereby reducing the need to physically grow plants and wait for observable results. Additionally, some symptoms are difficult to reproduce in certain species but can be learned from others, making generative models a practical solution for simulating rare or hard-to-cultivate cases.

Keywords: Generative Models, Diffusion, Domain Transfer.

Project 2: This project explores multimodal representation learning for fungal microscopic images under various treatments, using vision-language models (CLIP and its variants). We approached the problem from two directions: (1) optimizing a shared representation space that balances effectiveness with interpretability, and (2) refining textual descriptions to better align with domain-specific terminology, addressing the limitations of CLIP's understanding in this context.

Keywords: Biological Imaging, Multimodal Representation Learning, Vision-Language Models, Explainability, OOD detection

Researcher – CLINC 5.1

Oct. 2021 – Jun. 2023

Worked on multimodal representation learning for urological tumour data (CT, MRI, histopathology) using SOTA methods under limited data conditions. Collaborated with clinical and academic partners at Heidelberg University to validate results and align models with diagnostic requirements.

Keywords: Medical Imaging, Multimodal Representation Learning

Research Assistant in Computer Vision Lab, Heidelberg University

Aug. 2019 – Sep. 2020

Collaborated with Artsiom Sanakoyeu and Vadim Tschernezki

Advised by Prof. Björn Ommer

Topic: Improving Deep Metric Learning by Divide and Conquer. I co-developed a progressive training framework to improve generalization in deep metric learning. Conducted extensive experiments and ablation studies on six benchmark datasets. This work was later formalized as my Master's thesis and got accepted in TPAMI (2021).

Keywords: Deep Metric Learning, Deep Metric Learning, Image Retrieval, Similarity Learning, Representation Learning.

Research Internship in Computer Vision Lab, Heidelberg University

Aug. 2018 – Nov. 2018

Collaborated with Dmytro Kotovenko and Artsiom Sanakoyeu

Advised by Prof. Björn Ommer

Topic: A Content Transformation Block for Style Transfer. We proposed and developed a content-aware style transfer method that adapts style locally based on image content. Proposed and implemented both qualitative and quantitative evaluation strategies to validate the approach.

Keywords: VAE, GAN, Style Transfer, Content-Style Separation, Texture Synthesis.

SKILLS

Development skills: Python, DL Frameworks (PyTorch, HF Accelerate, DDP, Lightning), SLURM, Linux, AI coding

Language: English, fluent; German, B2 (intermediate); Mandarin and Cantonese, native speaker.

MISCELLANEOUS

- Kayaking, Hiking, and all other nature-related sports.
- Cooking and Beer Drinking.
- Gaming and Sci-fi Reading.